Inventory Demand Forecasting via Kalman Filtering and Global Gradient Boosting

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Phase 1 Machine Learning Project

Abstract—Reliable demand forecasts help retailers plan stock while avoiding excess inventory and missed sales. This Phase 1 study compares three understandable machine-learning approaches on the M5 Forecasting Accuracy data: a seven-day Seasonal Naive baseline, one local-level Kalman filter per product-store series, and one global histogram gradient-boosting model. We selected 100 series reproducibly, with 10 series from each of the ten stores, and evaluated four consecutive seven-day forecast windows using rolling origins. All models predicted the same 2,800 series-date-window points. Gradient Boosting produced the best overall point forecasts, with MAE 1.0224, RMSE 2.0397, and WAPE 67.03%. The Kalman filter was close, with MAE 1.0595, and supplied approximate Gaussian uncertainty intervals. Its mean pinball loss was 0.3413 and its nominal 80% interval covered 87.57% of observations, indicating conservative intervals on this sample. These results support the value of shared lag and calendar relationships, but apply only to this limited experiment and do not establish production readiness or full M5 performance.

Index Terms—demand forecasting, retail, Kalman filter, gradient boosting, rolling-origin evaluation, M5

I. Introduction and Problem Statement

Retailers need forecasts at product and store level to make replenishment and inventory decisions. Forecasting is difficult because daily sales can be sparse, seasonal, affected by calendar events, and different across products. Recorded sales are also not always identical to customer demand: a zero may mean no purchase, but it may also occur when an item is unavailable.

This project implements the machine-learning part of the assigned inventory forecasting problem. Its objectives are to: (1) build an interpretable weekly baseline; (2) estimate a changing local demand level and uncertainty with a Kalman filter; (3) learn shared nonlinear relationships with a global gradient-boosting model; and (4) compare all methods fairly with rolling forecast origins. DeepAR, LSTM, and hybrid models belong to future Phase 2 work and are not implemented here.

II. Related Work

The M5 competition provides hierarchical Walmart unit-sales series together with calendar and price information. Its published analysis highlights intermittent retail demand, explanatory variables, and the success of machine-learning approaches such as boosted trees [1]. A seasonal naive forecast remains useful because it is transparent: for weekly seasonality, it repeats the value observed seven days earlier [2].

TABLE I
Verified M5 input dimensions

File	Rows	Columns
Sales	30,490	1,919
Calendar	1,969	14
Selling prices	6,841,121	4

Kalman's state-space formulation recursively combines a model prediction with a noisy observation [3]. In a local-level model, the hidden state represents the current demand level, which can change over time. Gradient boosting instead adds decision trees sequentially so that later trees reduce errors left by earlier trees [4]. A global model can learn one set of relationships across many related series.

Forecast accuracy must be tested out of sample. Rolling-origin evaluation repeatedly moves the forecast date forward and avoids random splits that would mix past and future observations [5]. For probabilistic forecasts, quantile or pinball loss is a proper scoring approach, while interval coverage checks how frequently observations fall inside the reported range [6].

III. Dataset and Exploratory Analysis

We used the M5 Forecasting Accuracy files. Table I gives the verified dimensions. The sales table contains 30,490 product-store series and 1,913 daily columns, from d_1 to d_1913. The calendar maps these day codes to dates, weekdays, events, and state-level SNAP indicators. Selling prices are recorded by store, item, and week.

The initial quality checks found no complete duplicate rows, no missing sales values, no missing prices, and no gaps in the calendar dates used by sales. Calendar event fields are often blank because most dates have no named event. Across 58,327,370 daily sales observations, mean sales were 1.13 units, the median was zero, the maximum was 763, and 68.20% of observations were zero. Figure 1 shows the strongly right-skewed distribution with the display limited to the 99.5th percentile for readability. The original values were not changed.

Zero recorded sales were retained. The files do not directly identify stockouts, so zero sales cannot safely be relabelled as zero customer demand or as lost demand. Aggregate weekday averages also show a weekly pattern:

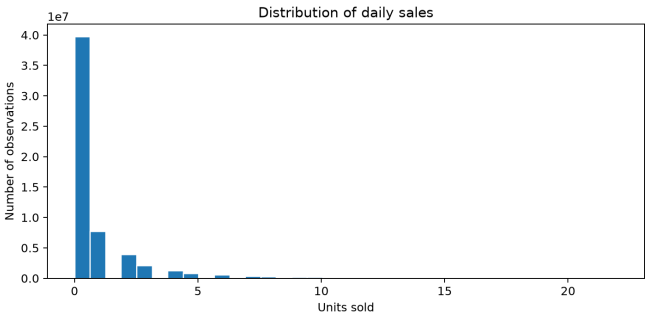


Fig. 1. Distribution of recorded daily unit sales.

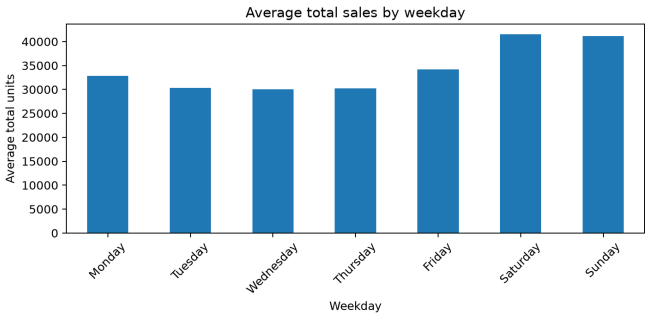


Fig. 2. Average total sales by weekday across the full sales file.

Saturday and Sunday averaged about 41,547 and 41,130 total units, compared with about 30,010 on Wednesday. Figure 2 summarizes this pattern.

Daily totals remain variable even after aggregation, while weekly and monthly sums make broader changes easier to see. Figure 3 presents these three time scales. The plots are descriptive summaries of the full dataset; model evaluation remains at individual product-store level.

IV. Feature Engineering

To keep the experiment understandable and fast, a fixed random seed selected 100 product-store series, balanced as 10 from each of the ten stores. Reshaping the selected histories produced 191,300 rows, exactly $100 \times 1,913$, with one row per series and day.

For series i on day t, the lag features are

$$x_{i,t}^{(k)} = y_{i,t-k}, \qquad k \in \{1, 7, 28\}, \tag{1}$$

where $y_{i,t}$ is recorded unit sales. The rolling means use only earlier observations:

$$\bar{y}_{i,t}^{(w)} = \frac{1}{w} \sum_{j=1}^{w} y_{i,t-j}, \qquad w \in \{7, 28\}. \tag{2}$$

Shifting before rolling is essential because including $y_{i,t}$ would leak the target into its own predictors.

The global model used 12 inputs: lag_1, lag_7, lag_28, rolling_mean_7, rolling_mean_28, weekday, month, weekend, primary event, SNAP, item code, and



Fig. 3. Aggregate daily, weekly, and monthly sales patterns used for exploratory analysis.

store code. Item and store codes are identifiers, not ordinal measurements. Selling price was inspected during EDA but was not used as a forecasting feature because this experiment did not assume future price availability.

V. Methodology

A. Rolling-Origin Design

The final 28 observed days were divided into four consecutive seven-day windows. The forecast origins were March 27, April 3, April 10, and April 17, 2016; their forecast periods ended on April 3, 10, 17, and 24, respectively. At each origin, a model could use actual demand only through that origin. Actuals from a completed window became history for the next origin, matching repeated real forecasting decisions.

B. Seasonal Naive

The baseline predicts the matching weekday from the previous week:

$$\hat{y}_{i,t} = y_{i,t-7}. \tag{3}$$

Because the horizon is seven days, every source value is known at the forecast origin.

C. Local-Level Kalman Filter

One scalar filter was fitted independently to each series at every origin. The local-level equations are

$$\ell_t = \ell_{t-1} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, Q), \tag{4}$$

$$y_t = \ell_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, R). \tag{5}$$

The implementation estimates Q and R from historical first differences, predicts the state variance, computes the Kalman gain, and updates the level after each available observation. The final filtered level is the point forecast. For horizon h, approximate 10th and 90th percentiles

are formed as $\ell \pm 1.2816\sqrt{P + hQ + R}$ and clipped at zero. This Gaussian interval is a simple approximation for nonnegative, intermittent count data; it is not separately calibrated.

D. Global Gradient Boosting

A histogram gradient-boosting regressor was trained across all selected series at each origin. It used squared-error loss, 80 boosting iterations, learning rate 0.08, at most 15 leaf nodes, minimum leaf size 30, L2 regularization 1.0, and random seed 42. Hyperparameters were fixed rather than tuned.

Forecasting was recursive. After predicting day one of a window, the model appended that prediction to the working history. Day two features therefore used the earlier prediction where required, and this continued through day seven. Hidden actual sales from the window were never used to construct its predictions.

VI. Evaluation Metrics

For n forecast points, the point metrics were

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n} |y_j - \hat{y}_j|, \tag{6}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n} (y_j - \hat{y}_j)^2}, \tag{7}$$

$$\mathrm{WAPE} = 100\frac{\sum_j |y_j - \hat{y}_j|}{\sum_j |y_j|}. \tag{8}$$

Lower values are better. RMSE penalizes large errors more strongly than MAE. In the implementation, WAPE is returned as undefined if the denominator is zero; this did not occur for the combined evaluation sets.

For quantile τ, pinball loss is

$$L_\tau(y, q) = \max\{\tau(y - q), (\tau - 1)(y - q)\}. \tag{9}$$

We averaged loss at $\tau = 0.1, 0.5, 0.9$ for the Kalman forecasts. The reported interval coverage is the percentage of actuals between $q_{0.1}$ and $q_{0.9}$. An 80% interval is ideally close to 80% coverage over a suitable sample; higher coverage can indicate intervals that are too wide.

VII. Results and Discussion

Each model generated 2,800 forecasts: 100 series, seven dates, and four windows. Table II reports the saved overall results. Gradient Boosting ranked first on all point metrics. Relative to Seasonal Naive, its MAE was about 15.5% lower. The Kalman filter was second and added probabilistic information not supplied by the two point-only models.

Table III and Fig. 4 show that Gradient Boosting had the lowest MAE in each window. Error rose in windows 3 and 4 for all three methods, especially for Seasonal Naive. This suggests that those weeks were harder to forecast rather than that one model alone became unstable. The global model could combine short and weekly lags, rolling levels, calendar variables, and identity information, which likely explains its advantage on this sample. This is an experimental explanation, not evidence of universal superiority.

TABLE II
Overall rolling-origin results

Model	Points	MAE	RMSE	WAPE
Gradient Boosting	2,800	1.0224	2.0397	67.03%
Kalman Filter	2,800	1.0595	2.1752	69.46%
Seasonal Naive	2,800	1.2096	2.7084	79.30%

Kalman only: mean pinball loss = 0.3413; approximate 80% interval coverage = 87.57%.

TABLE III
MAE by seven-day forecast window

Window	Gradient Boosting	Kalman	Seasonal Naive
1	0.9617	1.0349	1.0986
2	0.9878	0.9901	1.1214
3	1.0782	1.1301	1.3043
4	1.0619	1.0830	1.3143

The Kalman mean pinball loss was 0.3413. Its 10th-to-90th percentile interval covered 87.57% of the observations, above the nominal 80% level. On this sample, the intervals were therefore conservative. Coverage alone does not establish full calibration, and the Gaussian assumption is imperfect for intermittent nonnegative sales.

VIII. Leakage Prevention and Reproducibility

The implementation includes several checks. Lag and rolling features are shifted and independently compared with earlier sales. Seasonal Naive explicitly uses $t - 7$. Kalman histories end at the current origin. Gradient Boosting training rows stop at the origin, while recursive steps use predictions rather than future actuals. Forecasts are generated before actual values are attached for evaluation. Assertions confirm equal forecast keys, equal actual values, no duplicate keys, no missing forecasts, nonnegative predictions, and exactly 2,800 points per model.

To reproduce the experiment, download calendar.csv, sell_prices.csv, and sales_train_validation.csv from the M5 Forecasting Accuracy competition page and place them in Datasets/raw/. Raw data are local and are not included in the repository. Install the packages in requirements.txt, then execute the three notebooks in order:

1) 01_data_exploration.ipynb,
2) 02_feature_engineering.ipynb,
3) 03_model_development.ipynb.

The second notebook saves the selected series, prepared table, configuration, and windows. The third regenerates predictions, result tables, and report figures. The scalar

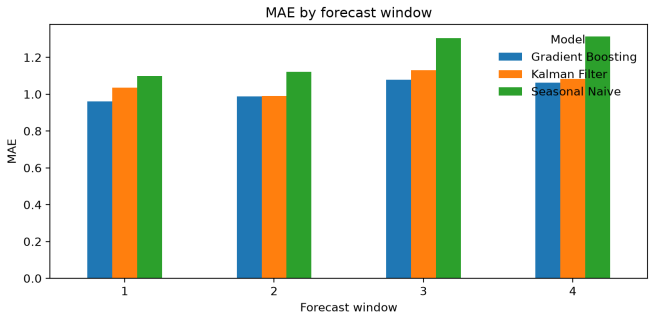


Fig. 4. Point-forecast MAE across the four rolling windows.

Kalman recursions are implemented directly, so no separate filtering package is required.

IX. Limitations and Future Work

The main limitations are as follows. Only 100 of the 30,490 series and four forecast origins were evaluated. Stockout information is not directly available. Hyperparameters were fixed instead of selected with nested validation. Only the primary event field was encoded. Numeric item and store codes are labels even though tree splits operate on their numeric values. The reported WAPE is an unweighted project metric, not the official M5 WRMSSE. Finally, the Kalman quantiles use a Gaussian approximation and are not guaranteed to be calibrated.

Future work should expand the number of series and origins, use explicit availability data if obtainable, test better categorical encoding and tuned baselines, and add Phase 2 global DeepAR or LSTM quantile forecasts on exactly the same evaluation keys. A hybrid model should be claimed only if an ablation demonstrates a real contribution.

X. Conclusion

This Phase 1 project built and fairly compared a transparent weekly baseline, a local state-space model, and a global tree model for retail demand. Gradient Boosting was the strongest point forecaster in this 100-series, four-window experiment, achieving MAE 1.0224, RMSE 2.0397, and WAPE 67.03%. The Kalman filter remained competitive and added approximate uncertainty intervals, though their 87.57% coverage was conservative relative to the nominal 80% level. The experiment demonstrates how past-only features, recursive forecasting, common forecast keys, and rolling origins can support a clear and leakage-aware comparison. It does not claim universal superiority or production readiness.

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